

# What's new in JupyterLab 4.6

and the tooling behind it

prototype

lint

build

test

Mike Krassowski



[github.com/krassowski/mini-summit](https://github.com/krassowski/mini-summit)

The screenshot displays the JupyterLab 4.6 interface. On the left, a file explorer shows a directory structure with files like 'Lorenz.ipynb' and 'study.csv'. The main area is split into two panes. The left pane shows a code editor for 'Lorenz.ipynb' with a title 'The Lorenz Differential Equations'. It contains a text block about the Lorenz system, a code cell with imports for 'matplotlib' and 'ipywidgets', and a text block titled 'Equations' showing the Lorenz equations. Below that is an 'Interactive plot' section with a code cell that uses 'solve\_lorenz' and 'interactive' to create a plot. The right pane shows a table viewer for 'study.csv' with columns 'subject', 'group', 'score', and 'minutes'. The table contains 20 rows of data.

|    | subject | group | score | minutes |
|----|---------|-------|-------|---------|
| 1  | S01     | A     | 71.2  | 34      |
| 2  | S02     | B     | 64.8  | 41      |
| 3  | S03     | A     | 88.1  | 29      |
| 4  | S04     | B     | 59.3  | 45      |
| 5  | S05     | A     | 77.6  | 31      |
| 6  | S06     | B     | 69.0  | 38      |
| 7  | S07     | A     | 92.4  | 27      |
| 8  | S08     | B     | 61.7  | 44      |
| 9  | S09     | A     | 80.3  | 33      |
| 10 | S10     | B     | 73.5  | 36      |
| 11 | S11     | A     | 85.9  | 30      |
| 12 | S12     | B     | 66.2  | 40      |
| 13 | S13     | A     | 79.8  | 32      |
| 14 | S14     | B     | 70.1  | 39      |
| 15 | S15     | A     | 83.0  | 28      |
| 16 | S16     | B     | 62.5  | 43      |
| 17 | S17     | A     | 90.7  | 26      |
| 18 | S18     | B     | 68.4  | 37      |
| 19 | S19     | A     | 75.2  | 35      |
| 20 | S20     | B     | 63.9  | 42      |

| Name                                                                                                  | Modified |
|-------------------------------------------------------------------------------------------------------|----------|
| data                                                                                                  | 11m ago  |
| experiments                                                                                           | 19m ago  |
| tests                                                                                                 | 42m ago  |
|  Lorenz.ipynb          | 36s ago  |
|  lorenz.py           | 2y ago   |
|  Data cleaning.ipynb | 18m ago  |
|  Analysis.ipynb      | 39m ago  |
|  README.md           | 2d ago   |

## Put the activity bar on any side

Left, right, top or bottom: **Settings › Interface › Activity Bar Position.**

1 of 8

[Back](#)

Next

Before we start, we import some preliminary libraries. We will also import (below) the accompanying `lorenz.py` file, which contains the actual simulation and plotting routine.

```
[1]: 1 %matplotlib inline
      2 from ipywidgets import interactive, fixed
```

# Equations

We explore the Lorenz system of differential equations:

$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = \rho x - y - xz$$

$$\dot{z} = -\beta z + xy$$

## Interactive plot

FileEditViewRunKernelTabsSettingsHelp

+

+

↑

↺

🔍

/ metanalysis /

| Name ^              | Modified |
|---------------------|----------|
| data                | 11m ago  |
| experiments         | 19m ago  |
| tests               | 42m ago  |
| Lorenz.ipynb        | 36s ago  |
| lorenz.py           | 2y ago   |
| Data cleaning.ipynb | 18m ago  |
| Analysis.ipynb      | 39m ago  |
| README.md           | 2d ago   |

Lorenz.ipynb

Python 3 (ipykernel)

# The Lorenz Differential Equations

Before we start, we import some preliminary libraries. We will also import (below) the accompanying `lorenz.py` file, which contains the actual solver and plotting routine.

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[1]: 1 %matplotlib inline
      2 from ipywidgets import interactive, fixed
```

## Equations

We explore the Lorenz system of differential equations:

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= \rho x - y - xz \\ \dot{z} &= -\beta z + xy\end{aligned}$$

## Interactive plot

Let's change  $(\sigma, \beta, \rho)$  with ipywidgets and examine the trajectories.

```
[2]: 1 from lorenz import solve_lorenz
      2 w = interactive(solve_lorenz, sigma=(0.0, 50.0), rho=(0.0, 50.0))
      3 w
```

## Histograms

```
[3]: 1 t, x_t = w.result
      2 xyz_avg = x_t.mean(axis=1)
      3 plot_histograms(xyz_avg)
```

The averages of x, y and z settle near the two attractors.

study.csv

Delimiter: ,

|    | subject | group | score | minutes |
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Go back to the cell you last edited

Two new buttons in the Table of Contents move back and forward through the cells you edited.

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BackNext

File

Edit

View

Run

Kernel

Tabs

Settings

Help

+

/ metanalysis / experiments /

Name

Modified

baseline.ipynb

2h ago

sweep-rho.ipynb

19m ago

sweep-sigma.ipynb

1h ago

{:}

params.json

1h ago

results.csv

19m ago

M

NOTES.md

3d ago

Lorenz.ipynb

+

Code

Python 3 (i

Type a path, press Tab

Click the breadcrumbs to edit the path. Tab completes folder names.

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Back

Next

ential Equations

We explore the preliminary libraries. We will also i

(below) the accompanying `lorenz.py` file, which contains the actual

and plotting routine.

[1]:

1 %matplotlib inline

2 **from** ipywidgets **import** interactive, fixed

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$\dot{x} = \sigma(y - x)$

$\dot{y} = \rho x - y - xz$

$\dot{z} = -\beta z + xy$

Interactive plot

Lorenz.ipynb



Paused in Debugger



## Debugger buttons next to your code

A small panel with Continue, Step and Stop floats over the notebook. Sources now open in the main area.

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[Back](#)

[Next](#)

# The Lorenz Differential Equations

Before we start, we import some preliminary libraries. We will use (below) the accompanying `lorenz.py` file, which contains the solving and plotting routine.

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## Equations

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Let's change  $(\sigma, \beta, \rho)$  with ipywidgets and examine the trajectories.

```
[2]: 1 from lorenz import solve_lorenz
      2 w = interactive(solve_lorenz, sigma=(0.0, 50.0), rho=(0.0, 50.0))
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```

## Histograms

|    |     |   | core | minutes |
|----|-----|---|------|---------|
|    |     |   | 71.2 | 34      |
|    |     |   | 64.8 | 41      |
|    |     |   | 88.1 | 29      |
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Table of Contents ×

LORENZ.IPYNB



The Lorenz Differential Equations

Equations

Interactive plot

🕒 JUPYTERLAB 4.6 • RELEASED 18 JUNE 2026

# One of the largest minor releases

68

new features and enhancements

97

bug fixes

95

contributors, 39 of them first-time

38

documentation improvements

171

maintenance tasks, a record

Also in

📖 Notebook 7.6

🌐 JupyterLite 0.8

Among minor releases, only 3.1 merged more pull requests.

📖 NOTEBOOK 7.6

# A scratchpad console

Try code on the same kernel without adding cells to the notebook.

Ctrl + B

The image shows a JupyterLab interface with a notebook on the left and a scratchpad console on the right. The notebook, titled 'analysis', contains two code cells. The first cell defines 'months' and 'temperatures' lists. The second cell calculates the average temperature and finds the warmest month. The scratchpad console, titled 'Python 3 (ipykernel)', shows the execution of the second code cell, displaying the output: 'Average temperature: 22.0°C' and 'Warmest month: Jul'.

**Sensor data analysis**

Explore the monthly temperature readings collected by the weather station.

```
[1]: months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
    temperatures = [18.2, 19.1, 20.3, 21.8, 23.4, 24.9, 26.1, 25.7, 24.2, 22.0, 19.8, 18.5]
```

```
[2]: average = sum(temperatures) / len(temperatures)
    warmest = months[temperatures.index(max(temperatures))]
    print(f"Average temperature: {average:.1f}°C")
    print(f"Warmest month: {warmest}")
```

Average temperature: 22.0°C  
Warmest month: Jul

Python 3.14.0 (main, Nov 19 2025, 23:04:28) [Clang 21.1.4 ]  
Type 'copyright', 'credits' or 'license' for more information  
IPython 9.11.0 -- An enhanced Interactive Python. Type '?' for help.  
Tip: The `%timeit` magic has a `-o` flag, which returns the results, making it easy to plot. See `%timeit?`.

```
[3]: sorted(zip(temperatures, months), reverse=True)[:3]
```

```
[3]: [(26.1, 'Jul'), (25.7, 'Aug'), (24.9, 'Jun')]
```

Also in 7.6: “Close and Shut Down” asks before it stops the kernel.

🌟 ALSO IN 4.6

# And more



## Shortcuts for any command

Add them in the shortcut editor. No JSON needed.



## Find and Replace in the Edit menu

`Ctrl` `H`, and `Ctrl` `Y` to redo on Windows and Linux.



## Language packs from Settings

“Install more languages...” opens the Extension Manager.



## Ghost text with highlighting

Inline completions are coloured like code, at every cursor.



## Zoom one panel

`Ctrl` + mouse wheel. Other panels do not change.



## Split a tab four ways

Left, right, up or down.



## Keyboard and screen readers

The terminal no longer keeps the keyboard focus. Focus is easier to see.



## Open in Terminal

From any folder in the file browser.



## Fewer mistakes

JupyterLab asks before it deletes a cell. Keys 1 to 6 no longer turn code into markdown.

# Faster builds, fewer dependencies

**~5x**

**faster builds**

JupyterLab, Notebook and extensions moved from webpack to Rspack.



**Build an extension without installing JupyterLab**

The build tooling now lives in `jupyter-builder`.



**Jupyter-specific lint rules in the extension template**

`@jupyter/eslint-plugin` is on by default for new extensions.

# This talk is also a report, and a thank-you



## FUNDED

four community proposals,  
in the Foundation's first  
funding round



## DELIVERED

the four proposals, working  
together



## THE COMMUNITY

reviewed, triaged, tested and  
adopted the work, and  
reported problems



## TRAVEL

OpenTeams paid for my trip  
to this summit

PART 2

# How did we get here?

Four proposals funded by the Jupyter Foundation.



## Prototype

Plugin Playground



## Lint

@jupyter/eslint-plugin



## Build

jupyter-builder



## Test

UI tests

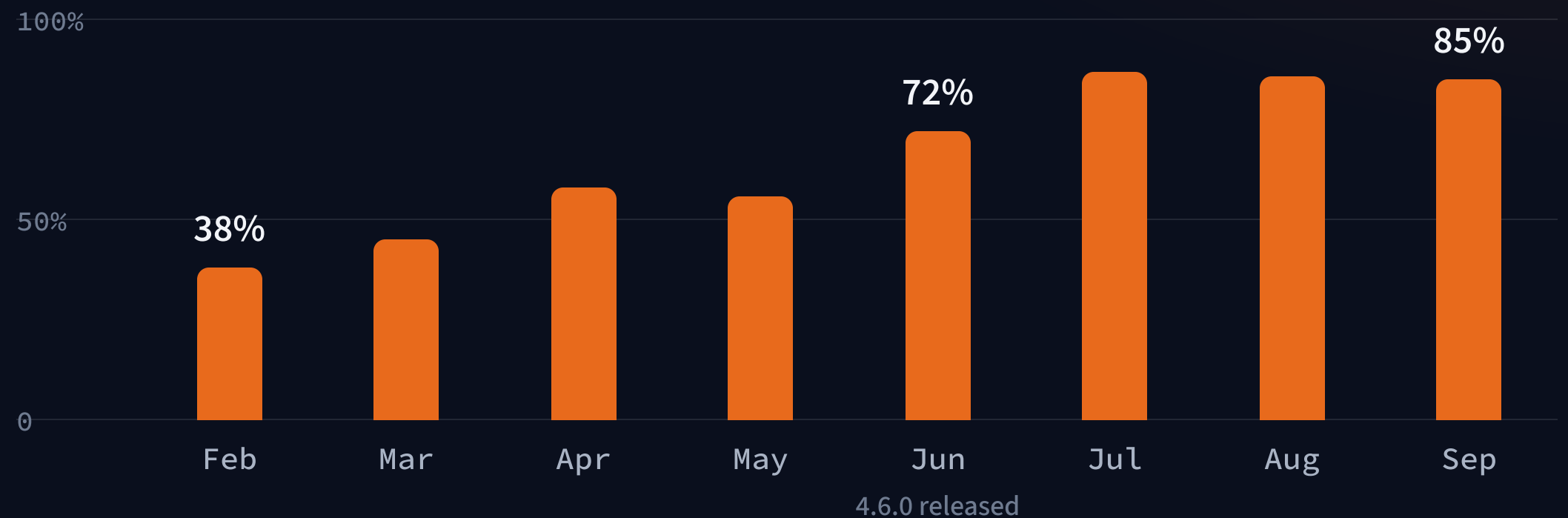
# Writing code became cheap

52%

of JupyterLab 4.6 pull requests that answered the AI question say **AI generated some of the content**

87 of 168 · the PR template asks since February 2026

Share of PRs with AI-generated content, by month opened



**28 of 30** merged PRs that changed a UI test from June to September were AI-assisted

**39** first-time contributors to JupyterLab 4.6

Still expensive:

🔧 prototyping

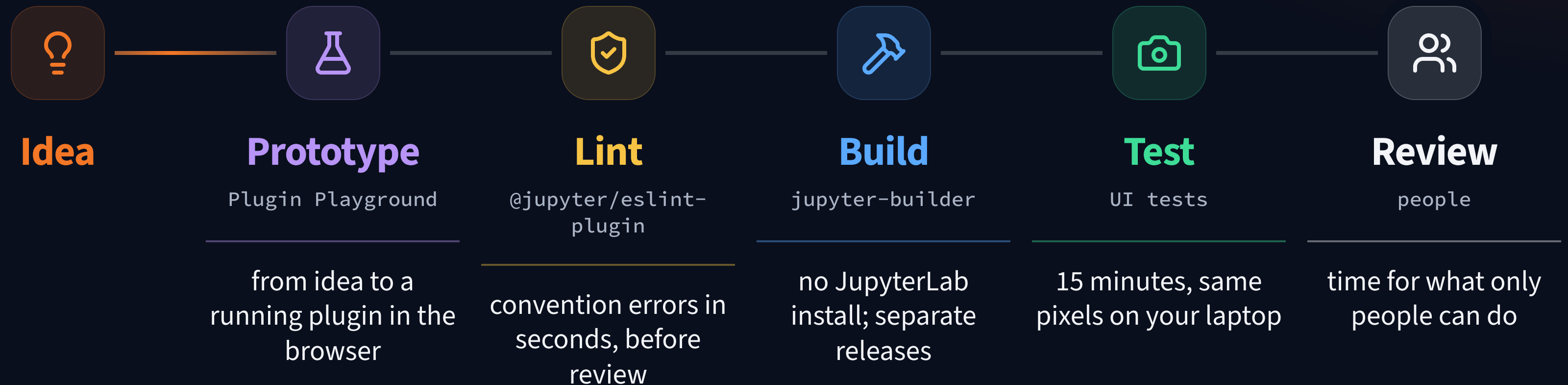
👥 reviewing

🔗 building

📷 testing

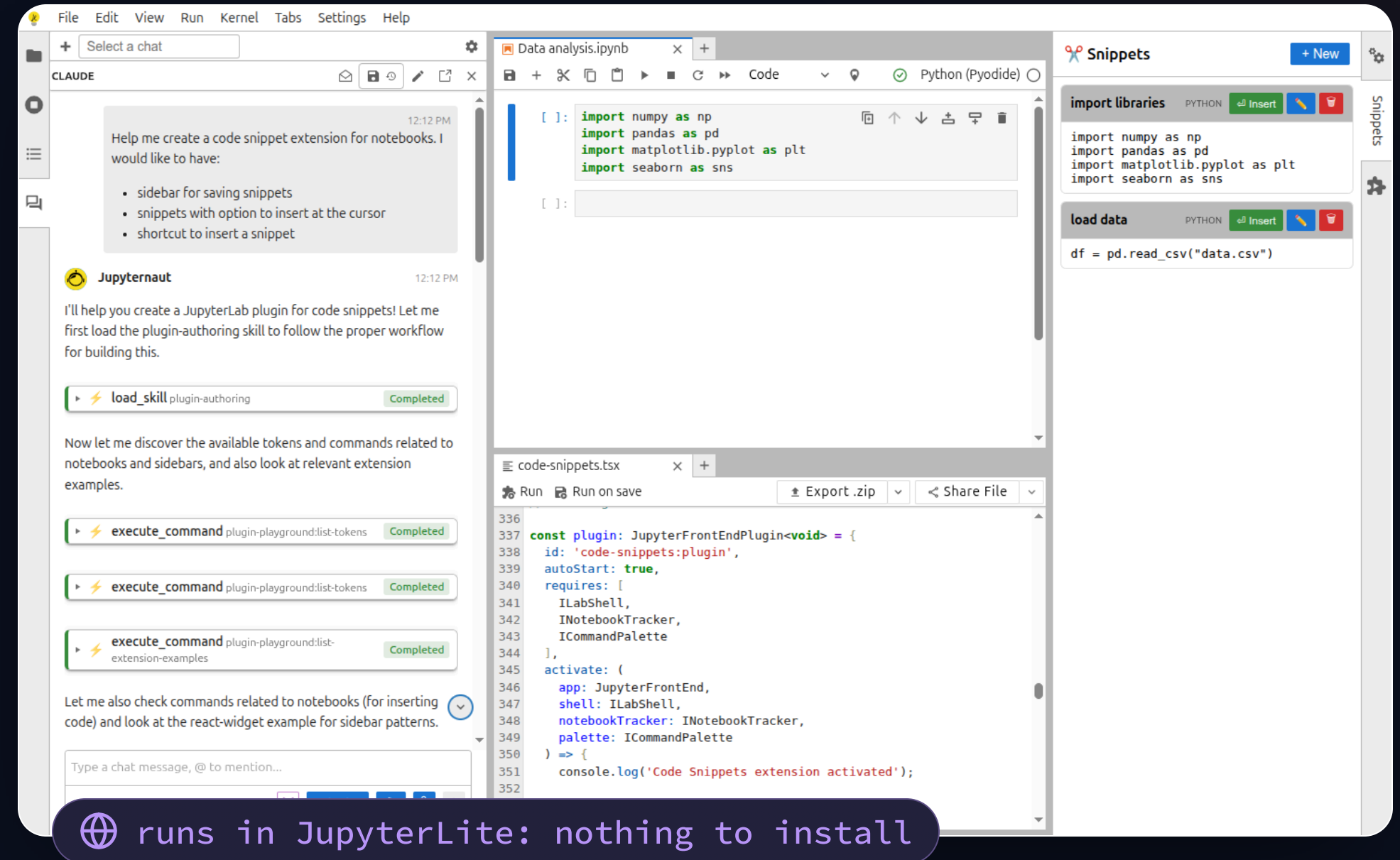
FROM IDEA TO RELEASE

# One pipeline, four proposals



# Plugin Playground 1.0

-  **Describe it** to the AI, or start from a file
-  **Insert** tokens and commands
-  **Load** the file as an extension, no build
-  **Run on save** while you iterate
-  **Share** a link for review
-  **Export** a wheel or a zip



# How the AI integration is built



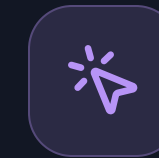
## No server needed

Built on `jupyterlite/ai`, so it works in JupyterLite and on Binder too.



## A guide for writing plugins

The AI follows written instructions, so first drafts have the structure that the playground expects.



## The same commands as you

The AI runs JupyterLab commands: create, load, export, share.

🔗 We sent four improvements to `jupyterlite/ai`. Next: code completion and error checking (LSP) in JupyterLite.

# @jupyter/eslint-plugin

8 → 23

rules, from 1.0 in May to 1.3 in September

Used in 50+ repositories, including

JupyterLab

Notebook

JupyterLite

extension template

jupyterlab-git

jupyter-collaboration

jupyter-chat

jupyterlab-execute-time

jupyterlab-notify



## Plugins

plugin-activation-args

4



## Translations

no-untranslated-string

4



## Memory leaks

require-signal-cleanup

5



## UI tests

galata-prefer-notebook-cell-helper

6



## Startup and deployment

prefer-lazy-imports

2



## Commands and settings

no-schema-enum

2

Led by  **Darshan Paudyal** (OpenTeams).

```
jnpm add -D @jupyter/eslint-plugin
```

# Agents can skip AGENTS.md. They cannot skip a lint error.

## AGENTS.md

- **68%** of instructions followed when there are 500 of them
- **+20%** cost, and no better results in general
- When an agent skips a line, nothing fails

Sometimes followed, and no error when it is not

## A lint rule

ui-tests/tests/run-cell.spec.ts

12:3 **error** Use `page.notebook.setCell()` / `runCell()`  
instead of raw cell selectors  
`jupyter/galata-prefer-notebook-cell-helper`

Runs every time. Names the file, the line and the fix.

**The agent does not need to know the rule. The linter tells it.**

# When should you write a custom lint rule?

- 1 TypeScript cannot check the convention
- 2 Niche framework, private codebase: not in training data
- 3 Breaking it causes no error, or a late one
- 4 A correct alternative can be suggested in the error message

Wrote the same review comment three times? **That comment is your rule.**

AGENTS.md

```
## Before you finish  
- Run `jlp eslint` and fix every  
  `jupyter/*` error.
```

# jupyter-builder

## BEFORE

 your extension

↓  
needs to build

 all of JupyterLab

server, application, every core package

↔ Packagers: `jupyterlab_pygments` needs JupyterLab to build,  
and JupyterLab includes it.

## NOW

 your extension

↓  
needs to build

 jupyter-builder

PyPI and npm



@jupyterlab/core-meta

compatibility metadata

```
- jupyter labextension build .  
+ jupyter-builder build . # --core-version 4.4.x
```

Led by  **Darshan Paudyal** (OpenTeams). Started in GSoC 2024 by Ronan Coutinho, mentored by Frédéric Collonval.

# A fix in 24 hours, without a JupyterLab release



## Rspack 2.0

The new bundler switched to Module Federation v2 by accident.



## Report

Extensions that use older shared packages break. The jupyter-chat maintainers report it.



## Under 24 hours

A jupyter-builder minor release goes back to Module Federation v1.



## For everyone else

No JupyterLab release needed. Extensions upgrade when they are ready.

The build system now has its own releases.

# UI tests you can trust

|                                  | BEFORE                   | NOW                      |
|----------------------------------|--------------------------|--------------------------|
| Full UI test run                 | 55 min                   | 15 min                   |
| New reference images             | 45 min, maintainers only | 1 min, anyone            |
| Flaky tests per run              | 17                       | 2.4                      |
| Runs with a hard failure         | 54%                      | 14%                      |
| Same pixels on your Linux laptop | ⊗ no                     | ✓ yes, with the CI fonts |

🐛 Fixing flaky tests found **19 real bugs** in JupyterLab. All are fixed.

# Agents need fast and reliable tests

55 min per run



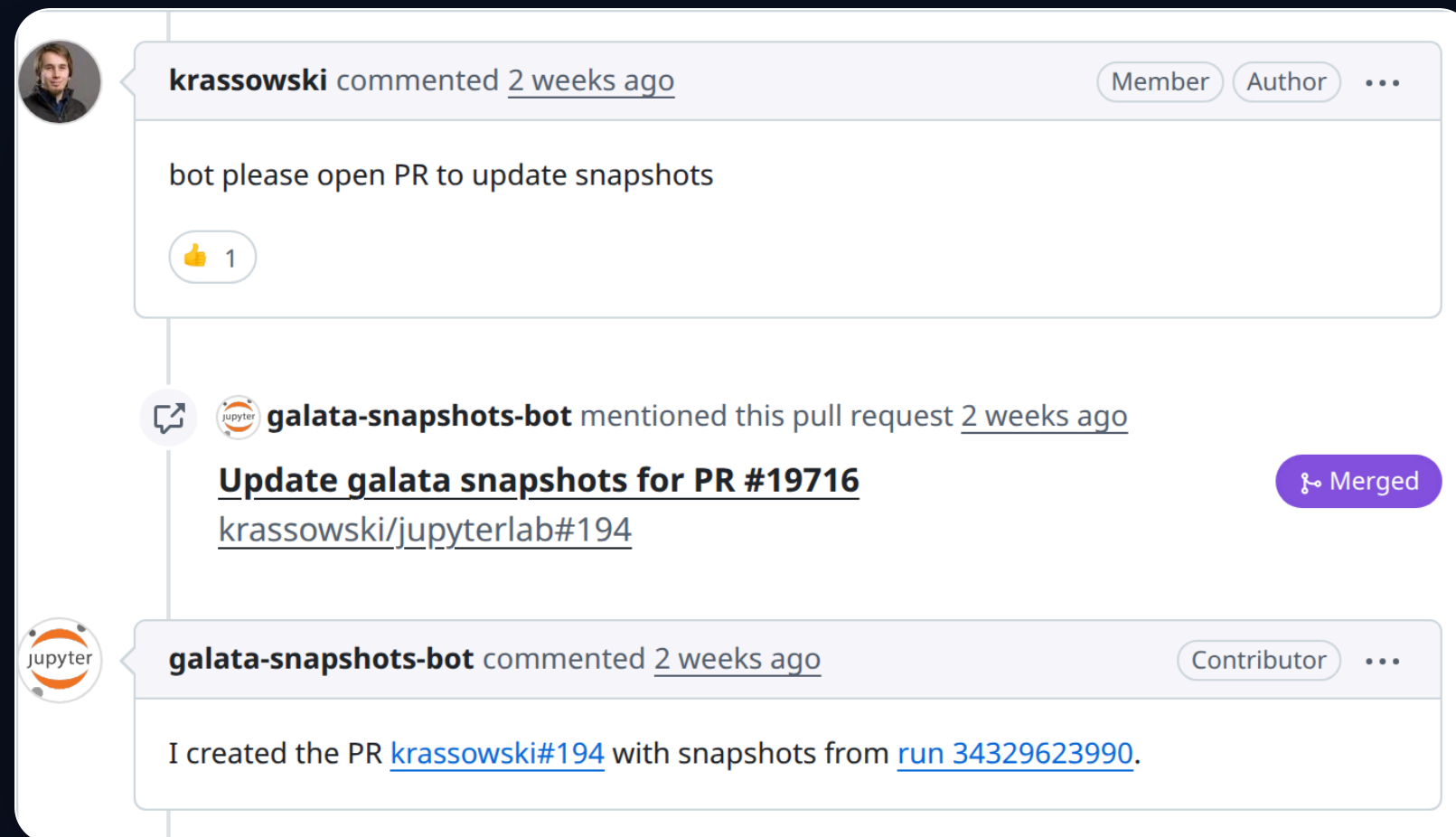
15 min per run



rounds of change and check in a working day

- ✓ A failure on your laptop is a real failure. No retries hide flaky tests, so agents stop adding fixed waits.
- ✓ The rules are in files that agents read: AGENTS.md, the contributing guide, lint rules.
- ✓ Every new or changed test has a video in the report.

# Anyone can update snapshots



**krassowski** commented [2 weeks ago](#) Member Author

bot please open PR to update snapshots

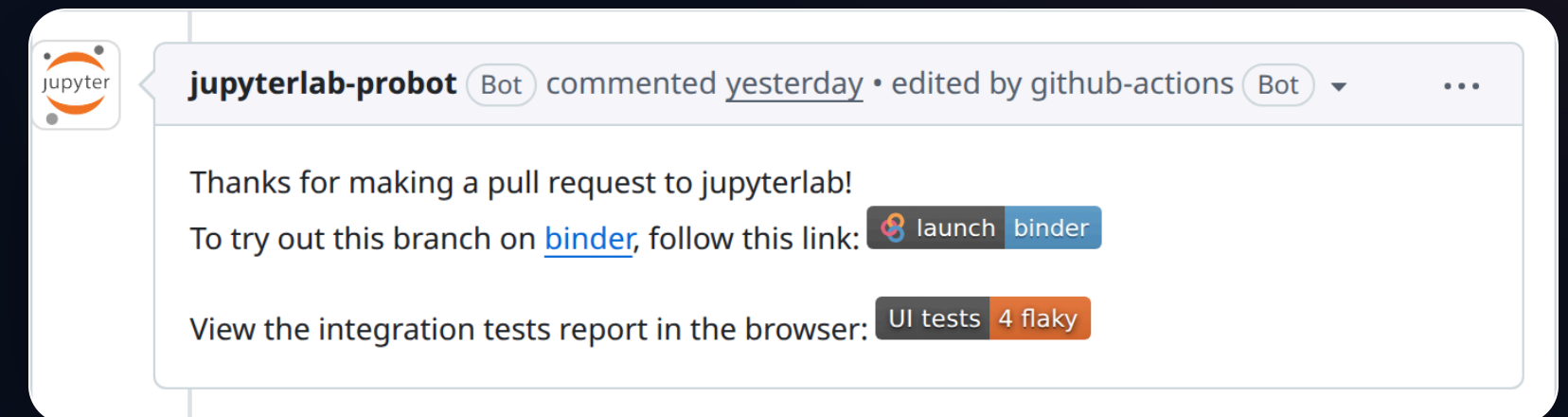
👍 1

**galata-snapshots-bot** mentioned this pull request [2 weeks ago](#)

**Update galata snapshots for PR #19716** 🔗 Merged  
[krassowski/jupyterlab#194](#)

**galata-snapshots-bot** commented [2 weeks ago](#) Contributor

I created the PR [krassowski#194](#) with snapshots from [run 34329623990](#).



**jupyterlab-probot** Bot commented [yesterday](#) • edited by github-actions Bot

Thanks for making a pull request to jupyterlab!

To try out this branch on [binder](#), follow this link: [launch binder](#)

View the integration tests report in the browser: [UI tests 4 flaky](#)

The badge opens the Playwright report in the browser. No download, no unzip.

Comment **please open PR to update snapshots**: a bot opens a PR with the new images in about a minute.



**Safe for pull requests from forks:** the bot commits to its own fork, a maintainer approves each run, and zizmor checks the workflow files.

# Use it in your project

## Plugin Playground

```
pip install jupyterlab-plugin-playground
```

Or open it in JupyterLite, no install.

## @jupyter/eslint-plugin

```
jlpm add -D @jupyter/eslint-plugin
```

Then add `configs.recommended`. Run it once on your extension.

## jupyter-builder

```
pip install jupyter-builder
```

`jupyter-builder build .` instead of `jupyter labextension build .`

## inline-playwright-report

```
jupyterlab/maintainer-tools
```

Turns a Playwright report into one HTML file that GitHub can show.

## ui-test-report-comment

```
jupyterlab/maintainer-tools
```

Posts the badge comment, with safety checks for forks.

## Stable screenshots

```
--disable-lcd-text
```

Pin fonts with `@fontsource`. Set `fullyParallel: true` so shards get equal work.

# New challenges with AI, and our answers

## CHALLENGE

## WHAT WE DO



**Code arrives faster than people can review it**



Move feedback from people to tools: lint, badges, snapshot PRs on request



**Agents skip written conventions**



Lint rules whose error message says how to fix it



**Flaky tests mislead agents**



Tests that give the same result on a laptop; a weekly report of flaky tests



**More first-time contributors**



Workflows that need no maintainer; an AI usage question in the PR template



**Bots that act on forks can be abused**



No write access in jobs that read fork content; maintainer approval; zizmor checks

# Four proposals, delivered



**Improve visual regression testing** #7

55 to 15 min; snapshots on request; local runs match CI

✓ COMPLETE



**Plugin Playground** #8

1.0 with AI; JupyterLite and Binder; share and export

✓ COMPLETE



**Finish the build system separation** #9

jupyter-builder 1.0 to 1.2; Rspack; no JupyterLab needed

✓ COMPLETE



**Custom linting** #11

23 rules; in the extension template; stricter JupyterLab CI

✓ COMPLETE

Delivered by



Reports: [jupyter-governance.github.io/funding-proposals](https://jupyter-governance.github.io/funding-proposals)

# Thank you

Questions?



Slides, sources and every link

[github.com/krassowski/mini-summit](https://github.com/krassowski/mini-summit)



## BUILT BY • OPENTEAMS



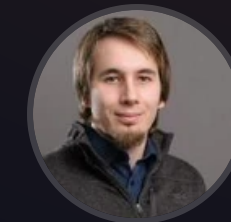
**Anuj Singh**

@MUFFANUJ  
Plugin Playground



**Darshan Paudyal**

@Darshan808  
lint, build



**Mike Krassowski**

@krassowski  
UI tests, guidance

## ADVICE AND SUPPORT • QUANSIGHT



**Smera Goel**

@smeragoel



**Matthias Bussonnier**

@Carreau



**Tania Allard**

@trallard

## ADMINISTRATIVE SUPPORT

Erika Oliphant (OpenTeams), Ashley Baal (Quansight)

## REVIEWS AND FEEDBACK

Jeremy Tuloup, Nicolas Brichet, Nicholas Bollweg, Jason Grout, Yann Pellegrini, Florence Haudin, Matt Fisher, Frédéric Collonval

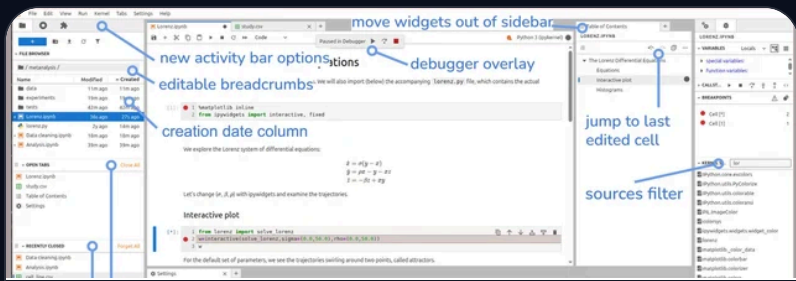
## AND

95 contributors to JupyterLab 4.6, the weekly triage group, and everyone who tried the pre-releases

 FURTHER READING

# Read more about the release and the tools

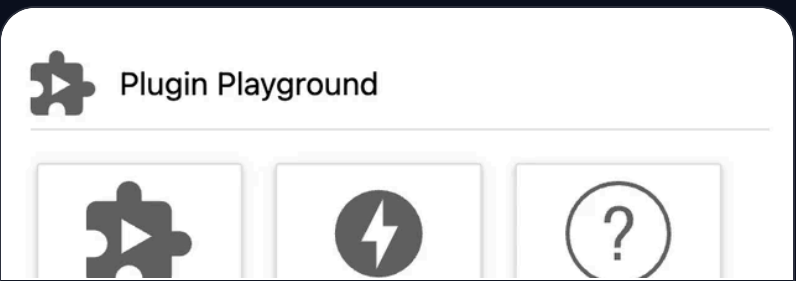
 JUPYTER BLOG



● RELEASE

**JupyterLab 4.6 and Notebook 7.6 are out!**

Michał Krassowski · 2 July 2026



● PROTOTYPE

**A User’s Journey with Plugin Playground: From First Idea to Installable JupyterLab Extension**

Anuj Singh · 28 May 2026



● LINT

**Catching Jupyter-specific bugs before CI does: announcing jupyter eslint plugin**

Darshan Paudyal · 11 June 2026



● BUILD

**Announcing jupyter-builder: A Standalone Build System for JupyterLab Extensions**

Darshan Paudyal · 19 June 2026

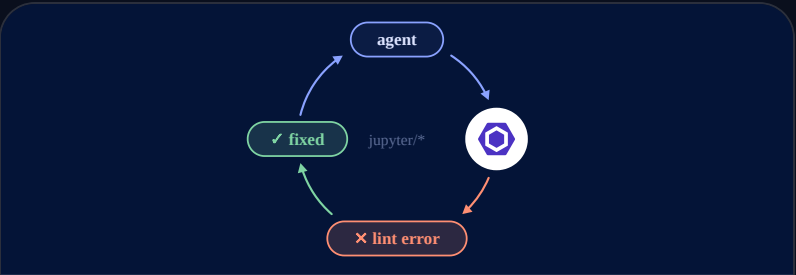
 OPENTEAMS ENGINEERING BLOG



● PROTOTYPE

**Plugin Playground AI Integration for Faster Plugin Prototyping**

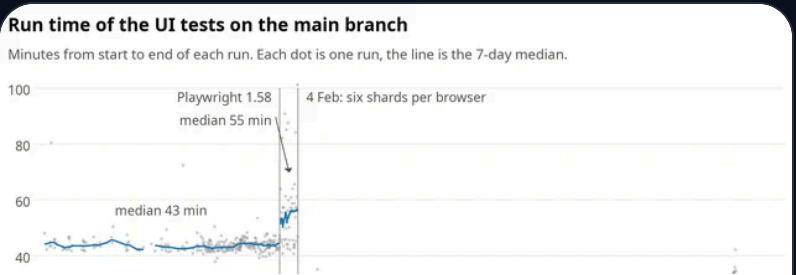
Anuj Singh · 24 April 2026



● LINT

**Your AI Agent Ignored AGENTS.md. Your Linter Won’t Let It.**

Darshan Paudyal · 1 October 2026



● TEST

**Reliable Visual Regression Testing for Humans and Coding Agents**

Mike Krassowski · October 2026



Every link, the slides and their sources

**[github.com/krassowski/mini-summit](https://github.com/krassowski/mini-summit)**

# AI usage in JupyterLab pull requests

| MONTH OPENED (2026)      | SAID YES | ANSWERED | SHARE | RELEASE                                                                                                                                                                                                                    | SAID YES | ANSWERED | SHARE |
|--------------------------|----------|----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-------|
| February (from the 13th) | 10       | 26       | 38%   | 4.6.0                                                                                                                                                                                                                      | 87       | 168      | 52%   |
| March                    | 19       | 42       | 45%   | After 4.6.0                                                                                                                                                                                                                | 87       | 106      | 82%   |
| April                    | 21       | 36       | 58%   | The question “Some or all of the content of this PR was generated by AI” is in the PR template since 13 February 2026. PRs by bots, and PRs that did not answer, are not counted. Script: <code>tools/ai_usage.py</code> . |          |          |       |
| May                      | 29       | 52       | 56%   |                                                                                                                                                                                                                            |          |          |       |
| June                     | 33       | 46       | 72%   |                                                                                                                                                                                                                            |          |          |       |
| July                     | 33       | 38       | 87%   |                                                                                                                                                                                                                            |          |          |       |
| August                   | 18       | 21       | 86%   |                                                                                                                                                                                                                            |          |          |       |
| September                | 11       | 13       | 85%   |                                                                                                                                                                                                                            |          |          |       |

# The 23 rules of @jupyter/eslint-plugin 1.3

## Plugins

plugin-activation-args  
plugin-description  
plugin-id-convention  
token-format

## Memory leaks

require-signal-cleanup  
require-signal-this-arg  
prefer-signal-this-arg  
require-disposable-ownership  
require-disposable-transfer

## Commands and settings

command-described-by  
no-schema-enum

## UI tests

galata-prefer-notebook-cell-helper  
galata-prefer-filebrowser-helper  
galata-prefer-menu-helper  
galata-prefer-context-menu-helper  
galata-prefer-sidebar-activity-helper  
require-soft-assertions-before-snapshots

## Translations

no-untranslated-string  
no-translation-concatenation  
no-dynamic-translation  
incorrect-translator-usage

## Startup and deployment

prefer-lazy-imports  
no-pageconfig-base-url

# Screenshots on other systems

|                                                     | BEFORE | NOW   |
|-----------------------------------------------------|--------|-------|
| Fedora 44, default fonts: screenshots that match CI | 1%     | 85%   |
| Fedora 44, default fonts: tests that pass           | 56%    | 93%   |
| Ubuntu 26.04 runner: screenshots that match CI      | 35%    | 100%  |
| Ubuntu 26.04 runner: tests that pass                | 71%    | 99.8% |

Measured on GitHub Actions on 26 September 2026. With the CI font packages installed, Fedora 44 and openSUSE Tumbleweed match all 255 comparisons. The remaining Fedora differences are Mermaid diagrams, which draw text with system fonts.